

Webinar Title

"Scaling Machine Learning with Spark MLlib: A Practical Introduction"

Time: 2PM(PT) to 2:45PM(PT)

ZOOM:

<https://ucidce.zoom.us/j/92173694932?pwd=c5CjTPN0LrCEMwM1wrljLF0h3Hk53g.1>

Meeting ID: 921 7369 4932

Passcode: 561752

Speaker: Jason Geng

Jason is the Chairman of the Board of Directors at the International Data Engineering and Science Association (IDEAS) and serves as the Chief Scientist of the IDEAS Incubator. He has over 20 years of experience in research and industry, specializing in big data, artificial intelligence, and data science. His primary research interests include cybersecurity, fintech, and blockchain applications. Jason has also served as an adjunct professor at the University of Southern California and California State University Long Beach.

Agenda (45 Minutes)

◆ 0:00 – 0:05 | Welcome & Introduction

- Brief intro to the speaker and webinar goals
- Importance of scalable machine learning
- Quick audience poll (e.g., “Have you used Spark or MLlib before?”)

◆ 0:05 – 0:15 | Overview of Apache Spark & MLlib

- What is Apache Spark?
- Core components: Spark Core, SQL, Streaming, MLlib, GraphX
- Why MLlib? (Scalability, distributed computing, ease of integration)
- Comparison with other ML frameworks (scikit-learn, TensorFlow, etc.)

◆ 0:15 – 0:25 | MLlib Core Concepts & Features

- Supported algorithms: classification, regression, clustering, collaborative filtering, etc.
- Pipelines and Transformers/Estimators (Spark ML API)
- DataFrames vs RDD-based MLlib
- Model training & evaluation
- Cross-validation & hyperparameter tuning

◆ 0:25 – 0:35 | Live Demo / Code Walkthrough

- Sample use case: Logistic regression or k-means clustering
- Loading and preprocessing data
- Training a model
- Evaluating and saving the model
- Running MLlib in local vs cluster mode (brief mention)

◆ 0:35 – 0:40 | Real-World Use Cases

- Industry examples (e.g., recommendation systems, fraud detection)
- Spark MLlib in cloud environments (Databricks, AWS EMR, GCP Dataproc)

◆ 0:40 – 0:45 | Q&A and Wrap-Up

- Open floor for audience questions
- Additional resources and learning paths
- Share feedback link / post-event survey